

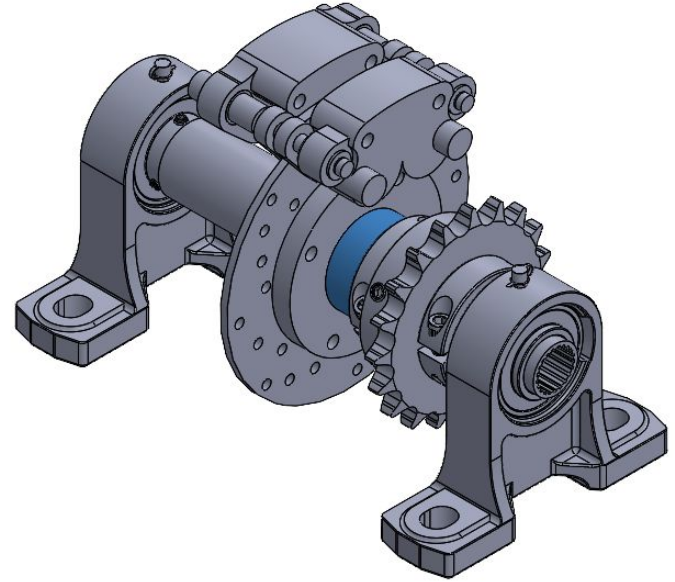
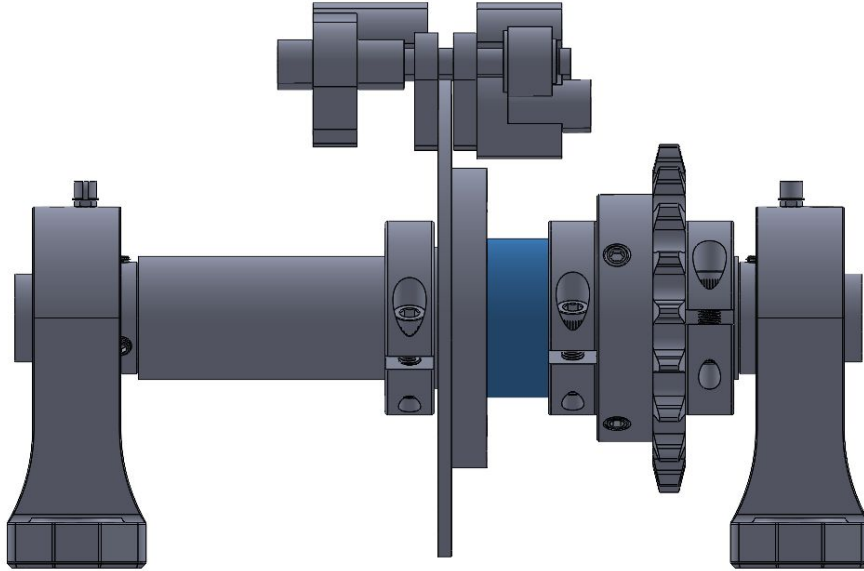
Rear Output

April 2025

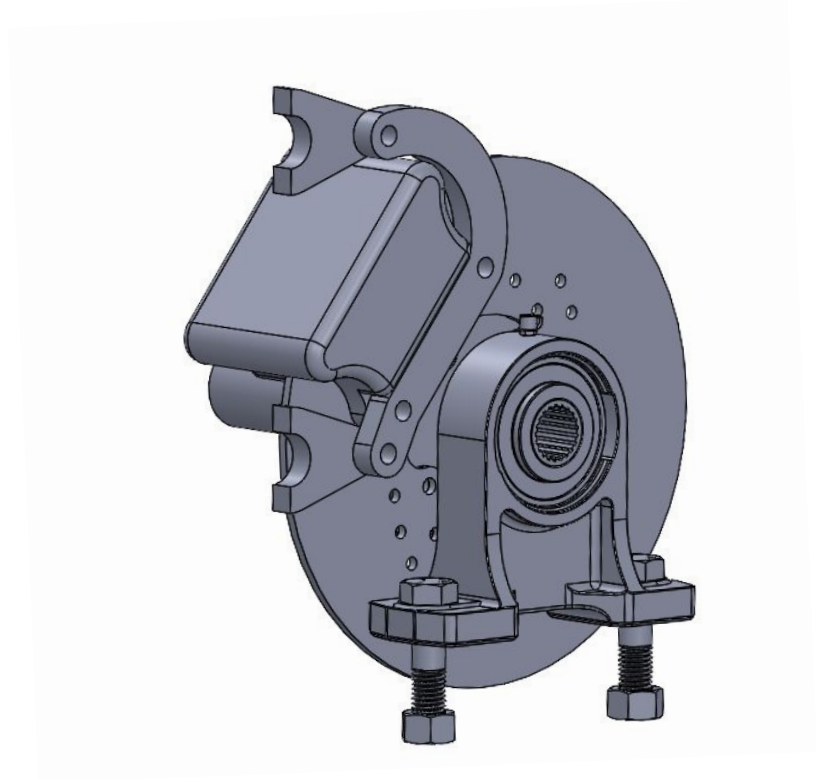
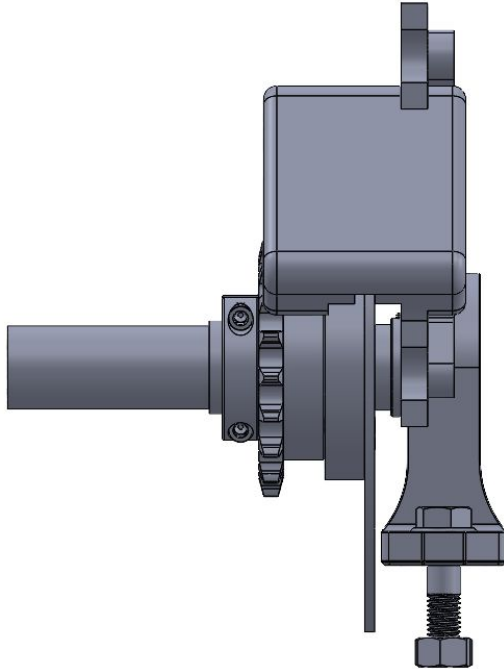
Heesung Han and Suwanee Li



CAD of Rear Output - last design review



CAD of New Rear Output

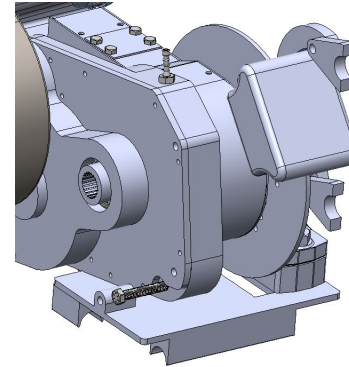
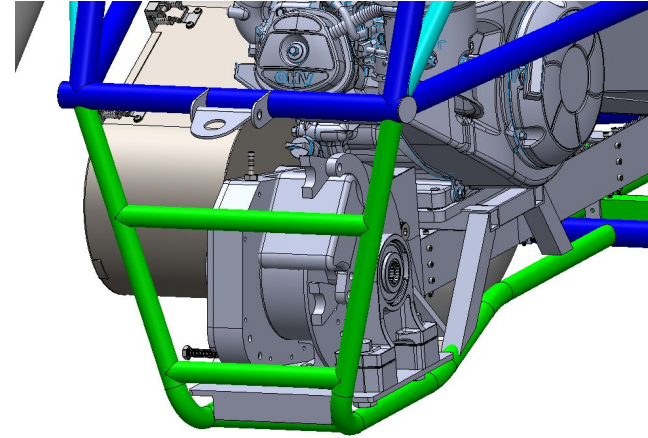


Changes made

- Removed pillow block near the gearbox side
- Changed order of components on rear spool
- Increased brake rotor size
- Added tabs and mounts for the brake caliper

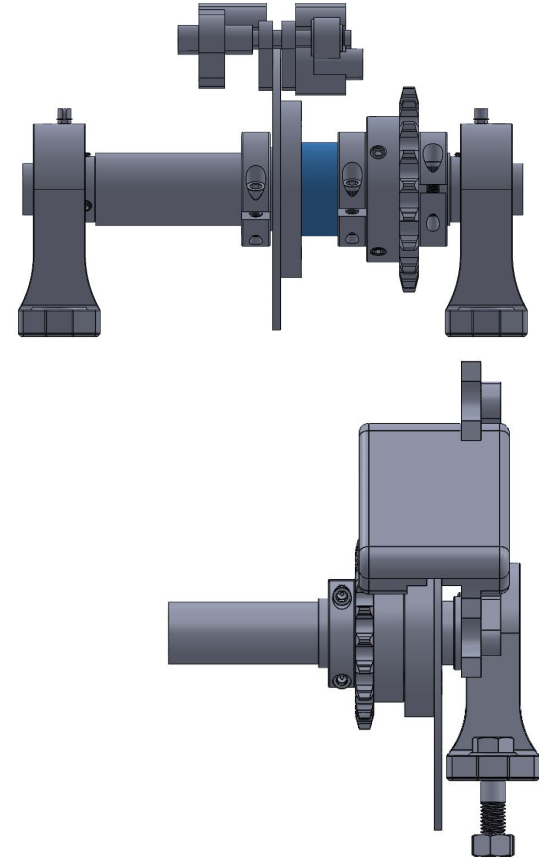
Pillow block removal

- Removal of the pillow block: took up too much space
- Now mounted to the gearbox with a plate for alignment



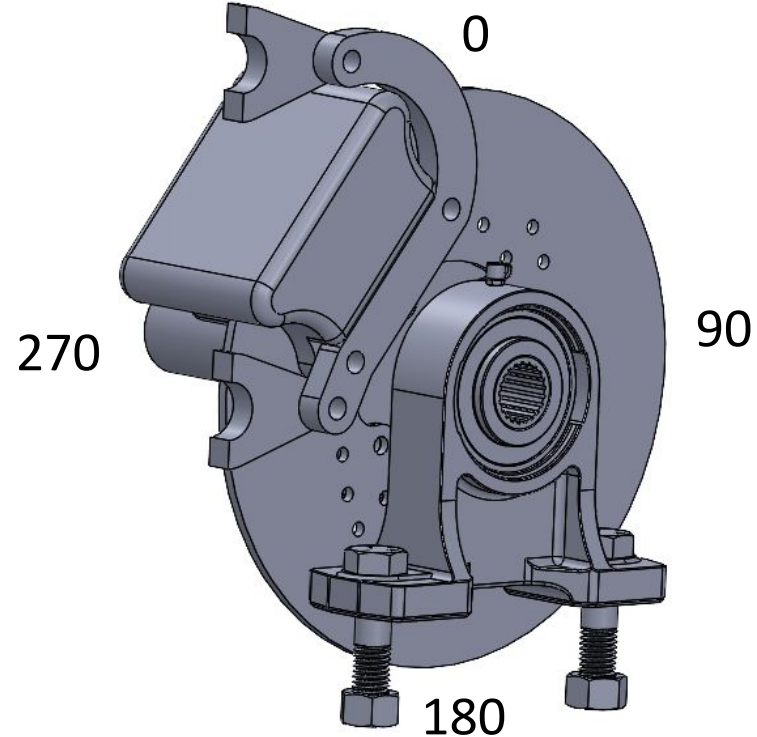
Order of components

- The old order: gearbox, brake rotor, than the sprocket.
- The new order: gearbox, sprocket, than brake rotor.
- We did this to center the sprocket near the middle.



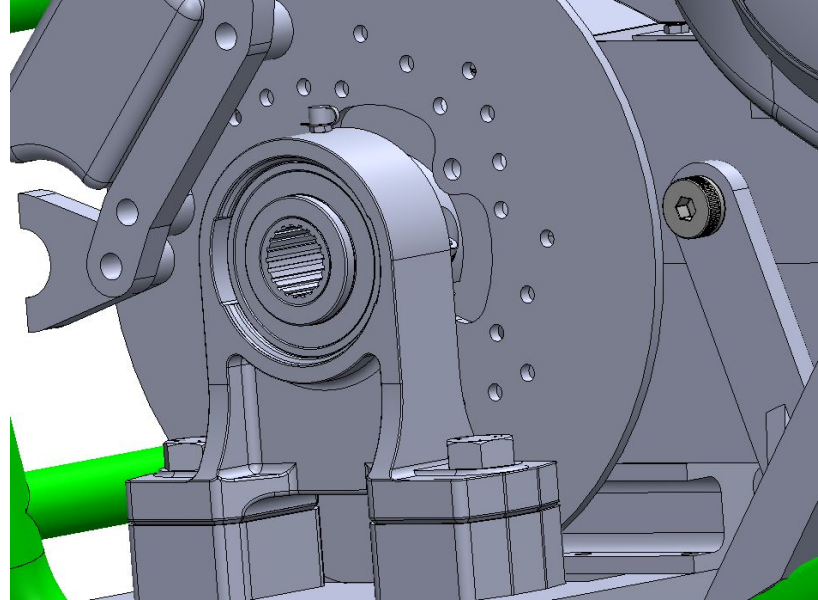
Brake Caliper Placement & Rotor Size

- 315° position to avoid contact with pillow blocks screws
- Brake rotor changed to 10 in diameter because Calipers support 10 in



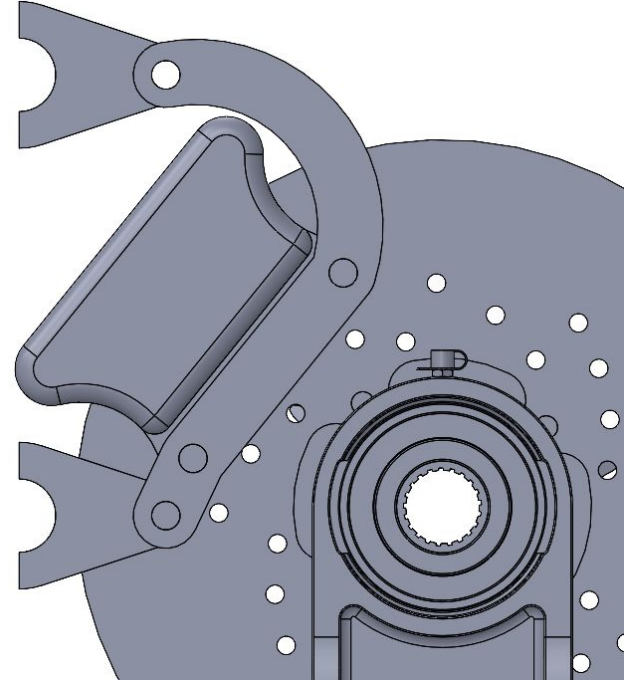
Flange

- Flange increased by 0.25 inches: preventing contact with chain tensioner standoff



Tabs and Mounts

- Tabs will be mounted to the back portion of the chassis.
- The mounts will screw into the tabs and the brake caliper.



Changes to Brake Caliper

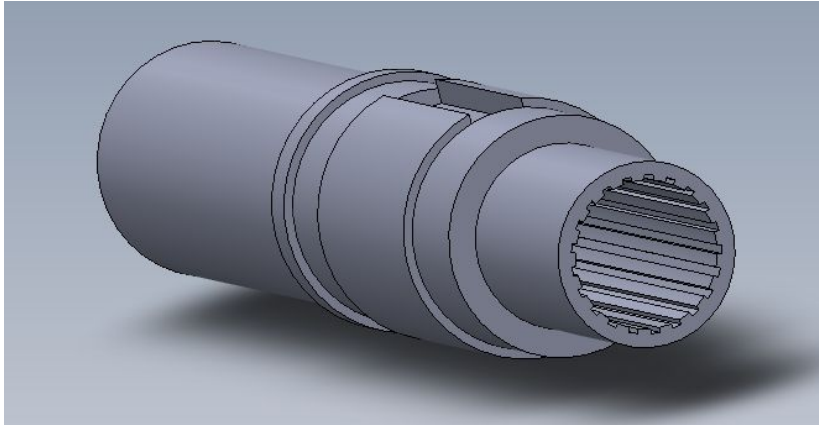
- Old Brake Caliper: 1 piston
- New Brake Caliper: 2 pistons
- For better clamping for brakes and also better for heat dissipation



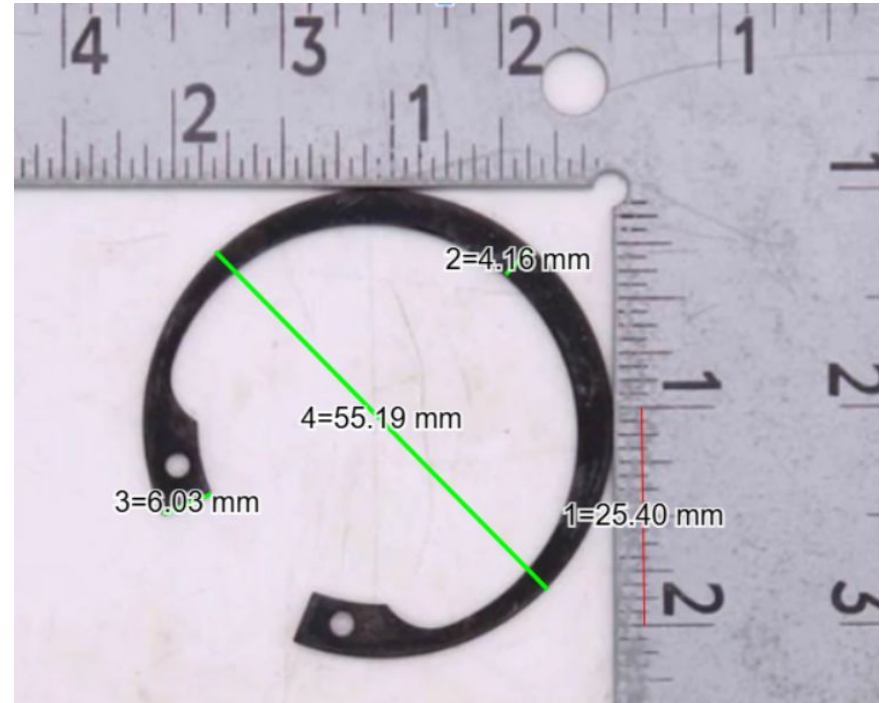
[Wilwood Caliper Link](#)

C-clips for Retaining half-shaft in Place

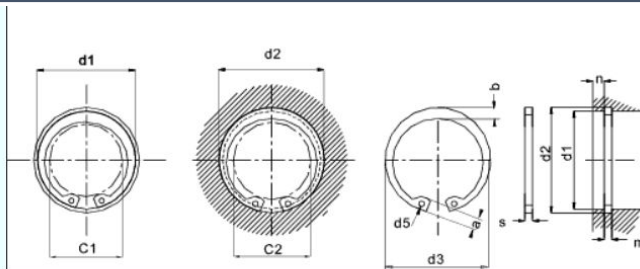
Cut groove in shaft -> To use an internal (bore) C-clip -> Half shafts stay in place
External (shaft)



[Compatible Arctic Cat](#)
[C-clips for Half Shaft Link](#)



Groove Dimensions



Nom size	Circlip Dimensions									Groove Dimensions					Groove Strength	Circlip Strength
d ₁	s	s (tol)	d ₃	d ₃ (tol)	a max	b	d ₅ min	C1	C2	d ₂	d ₂ (tol)	m min	t	n	F _n (kN)	F _r (kN)
50	2,00	-0,07	54,2	+1,10 -0,46	6,5	4,6	2,5	36,3	40,0	53,0	+0,30	2,15	1,50	4,5	40,50	60,8

	Measured	Chart
amax	6.03 mm	6.5 mm
b	4.16 mm	4.6 mm
d3	55.19 mm	54.2 mm

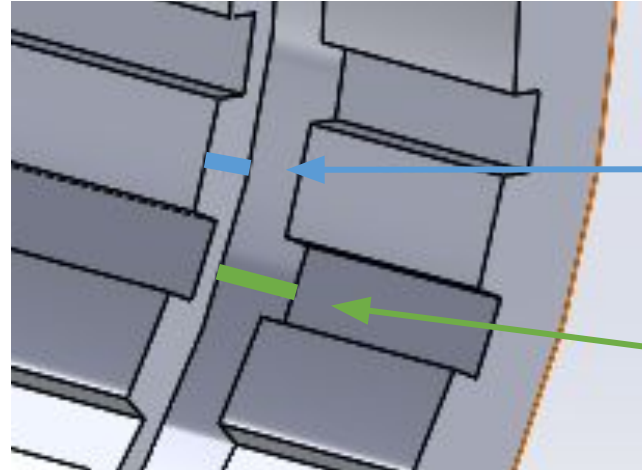
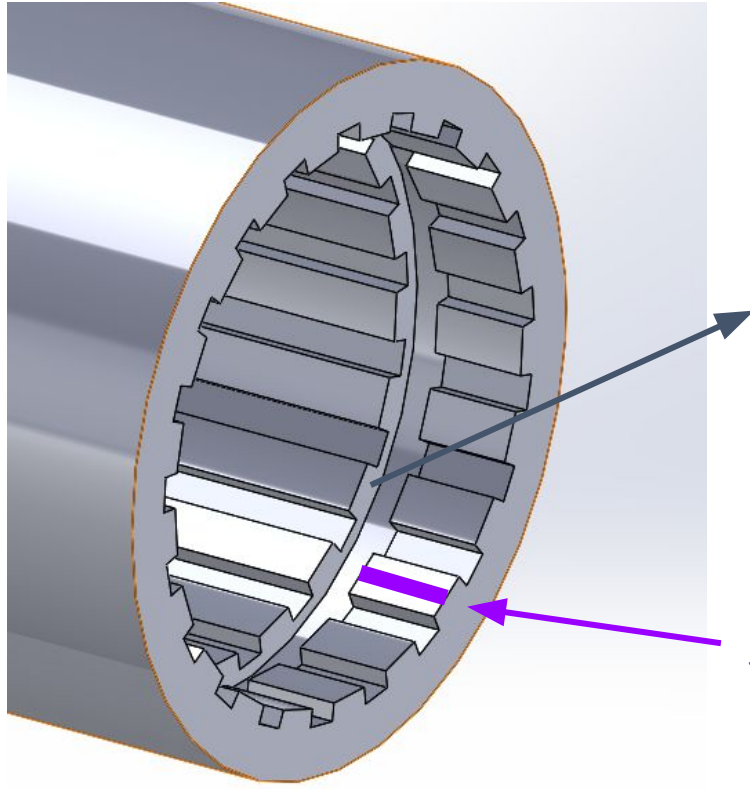
Deepness of cut is
 $(d_2 - d_1)/2 =$
 $(53-50)/2 = 1.5 \text{ mm}$

2.15 mm
wide cut

Cut 4.5 mm inwards
from the outside

Closest values found

Cut into shaft in CAD



Deepness 1.5 mm

2.15 mm wide cut

Cut 4.5 mm inwards
from the outside

Questions?

Cost (ADD TO THE BUDGET SHEET)



Brake Calipers = 145.45

Pillow Block =

C-clips = 4 shafts * 4 = 16

Miscellaneous =

- By now mounting the calipers and using a 2 piston brake calipers it has resulted it means the tolerance for the placement is now more specific

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